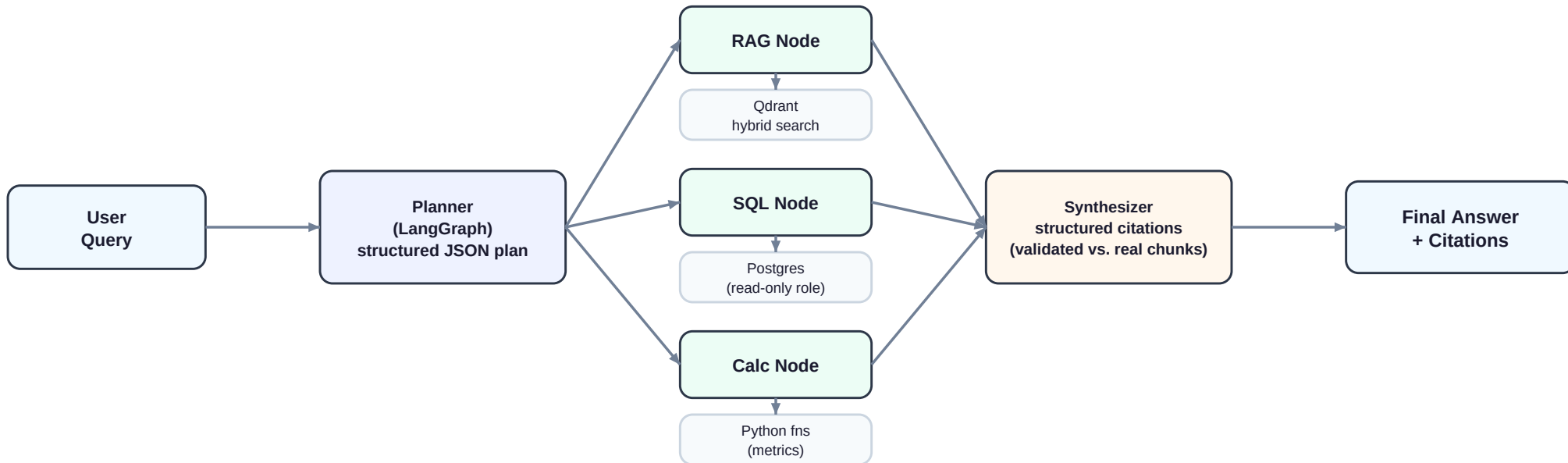


Financial Intelligence Agent — System Architecture

RAG + Text-to-SQL + Calculation agent, orchestrated with LangGraph, deployed via Docker Compose on a fully open-source stack.

github.com/vishwavidhan33-arch/Financial-Intelligence-Agent



Core Strategies

- Plan-then-execute orchestration (inspectable vs. opaque ReAct loops)
- Hybrid retrieval: dense + BM25 via Reciprocal Rank Fusion
- Hybrid chunking: structure-aware text + atomic table chunks
- Deterministic-first calculation; LLM tool-calling only for edge cases

Safety & Grounding

- sqlglot-validated SQL: single-statement, SELECT-only, allow-listed
- Read-only DB role as a second, independent line of defense
- Restricted AST evaluator for LLM formulas — no `eval()/exec()`
- Structured citations verified against real retrieved chunk IDs

Stack (fully open-source)

- Orchestration: LangGraph, Pydantic
- LLM serving: Ollama (Llama 3.1 / Qwen2.5 / Mistral)
- Retrieval: Docling, Qdrant, BGE embeddings + reranker, BM25
- Data: PostgreSQL, SQLAlchemy, sqlglot
- API & deploy: FastAPI, Docker Compose
- Evals: custom LLM-as-judge harness